Title: Week6 - 7 -Artificial Neural Network Implementation

Group Work: Students may work individually or in pairs (max 2 members per group**).**
Dataset: Kaggle — Android Permission Dataset
https://www.kaggle.com/datasets/saurabhshahane/android-permission-dataset

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Objectives / Intended Learning Outcomes:

By the end of this lab, you will be able to:

1. Acquire and load a real-world dataset into Python.
2. Clean and reformat the dataset by removing irrelevant attributes and handling missing values.
3. Explore the **class distribution** of benign vs. malware apps and deal with imbalance.
4. Train and evaluate models using **Artificial Neural Network**.
    - Identify the initial dimensionality of the input layer.
    - Create 4 more models with different number of hidden layers with units varies:
        1. ANN – model 1 => (6 layers) 128, 64,32,16, 8, 1
        2. ANN – Model 2 => (5 layers) 64,32,16, 8, 1
        3. ANN – Model 3 => (4 layers) 32,16, 8, 1
        4. ANN – Model 4 => (3 layers) 16, 8, 1
5. Compare model performance using metrics such as **Accuracy, Precision, Recall, F1-score** and **Confusion Matrix**.

Lab Workflow

Step 0: Setup

- Download the Kaggle dataset using either the Kaggle API or manual download.

Step 1: Load Data

- Inspect the dataset and identify the available columns.
- Output: List of column names

Step 2: Data Cleaning

- Remove unnecessary **non-numeric columns** (app names, descriptions, package IDs, etc.).
- Keep only **numeric features** and the **target label (Class)**.
- Columns with all zeros or constants add no value, drop them.
- Handle missing values and remove duplicates.
    - Output: List of column names
- Ensure the target is binary (`0 = Benign`, `1 = Malware`).

Step 3: Train/Test Split

- Split data into training and test sets using a **stratified split**.

Step 4: Modeling

- Conduct experiments in three phases of dataset preparation:
    1. Raw dataset (unprocessed numeric values).
    2. Normalized dataset (apply scaling such as MinMaxScaler or StandardScaler).
- For each dataset version, train the following ANN models with different hidden layer configurations:
    1. ANN – Model 1 (6 layers): 128 → 64 → 32 → 16 → 8 → 1
    2. ANN – Model 2 (5 layers): 64 → 32 → 16 → 8 → 1
    3. ANN – Model 3 (4 layers): 32 → 16 → 8 → 1
    4. ANN – Model 4 (3 layers): 16 → 8 → 1

Important: This means you will run 8 experiments in total (4 models × 2 dataset versions).

Step 5: Evaluation & Comparison

- Evaluate each trained model using the following performance metrics:
    - Accuracy
    - Precision
    - Recall
    - F1-score
- Generate a Confusion Matrix for at least one of the strongest models.
- Identify the best-performing model across all runs. Display its performance on both training and testing sets, and provide a discussion of results, including:
    - How raw vs. normalized vs. modified datasets affect performance.
    - Trade-offs between model complexity (number of layers/units) and evaluation metrics.